

Cyclic polynomials in Dirichlet-type Spaces of the unit bidisk

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Preliminaries

- $\mathbb{C}$ denotes the complex plane.
- $\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}$ is the open unit disk.
- $\mathbb{T} := \{z \in \mathbb{C} : |z| = 1\}$ is the unit circle.
- $\mathcal{B}(\mathcal{H})$ denotes the set of all bounded linear operators on a Hilbert space $\mathcal{H}$.
- Let M be a closed subspace of $\mathcal{H}$ and let $T \in \mathcal{B}(\mathcal{H})$. We say M is **T -invariant** if

$$T(M) \subseteq M.$$

- Given a function f , the set $\mathcal{Z}(f)$ denotes the zero set of f .

Cyclic Vector for a Single Operator

Let $T \in \mathcal{B}(\mathcal{H})$. A vector $x \in \mathcal{H}$ is called **cyclic for T** if

$$[x]_T := \overline{\text{span}}\{T^n x : n = 0, 1, 2, \dots\} = \mathcal{H}.$$

Equivalently,

$$\overline{\{p(T)x : p \text{ is a polynomial}\}} = \mathcal{H}.$$

In other words, the smallest closed T -invariant subspace containing x is $\mathcal{H}$.

Hardy Space $H^2(\mathbb{D})$

- The Hardy space $H^2(\mathbb{D})$ consists of all analytic functions in the unit disk

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

such that

$$\|f\|_{H^2}^2 := \sum_{n=0}^{\infty} |a_n|^2 < \infty.$$

- The shift operator $S : H^2(\mathbb{D}) \rightarrow H^2(\mathbb{D})$ is defined by

$$(Sf)(z) = zf(z).$$

Remark

- Polynomials are dense in $H^2(\mathbb{D})$.
- Monomials are orthogonal to each other in $H^2(\mathbb{D})$.

Cyclic Functions and Outer Functions

- A function $f \in H^2(\mathbb{D})$ is called *cyclic* if

$$[f] := \overline{\text{span}}\{z^n f : n \geq 0\} = H^2(\mathbb{D}).$$

- A consequence of Beurling's theorem yields that

$$f \text{ is cyclic in } H^2(\mathbb{D}) \iff f \text{ is outer,}$$

meaning

$$\log |f(0)| = \int_0^{2\pi} \log |f^*(e^{i\theta})| \frac{d\theta}{2\pi} \quad \text{and} \quad f(0) \neq 0.$$

Here $f^*(e^{i\theta}) = \lim_{r \rightarrow 1} f(re^{i\theta})$ which is known to exist for a.e. $\theta \in \mathbb{T}$ and for every $f \in H^2(\mathbb{D})$.

Dirichlet-Type Spaces in the Unit Disk

Let $\alpha \in \mathbb{R}$. The **Dirichlet-type space** D_α in the unit disk $\mathbb{D}$ consists of all holomorphic functions

$$f(z) = \sum_{n=0}^{\infty} a_n z^n, \quad z \in \mathbb{D},$$

such that

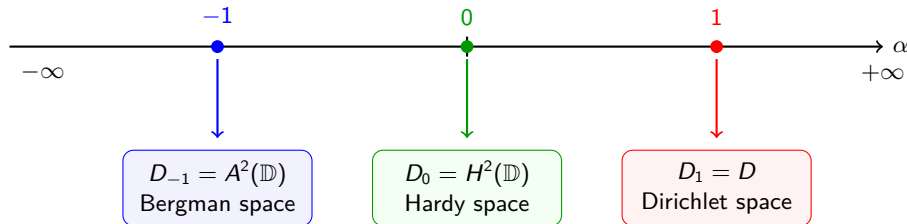
$$\|f\|_{D_\alpha}^2 := \sum_{n=0}^{\infty} (n+1)^\alpha |a_n|^2 < \infty.$$

Remark.

- D_α is a reproducing kernel Hilbert space with the inner product

$$\langle f, g \rangle_{D_\alpha} = \sum_{n=0}^{\infty} (n+1)^\alpha a_n \overline{b_n}, \quad f(z) = \sum_{n=0}^{\infty} a_n z^n, \quad g(z) = \sum_{n=0}^{\infty} b_n z^n.$$

The Scale of Dirichlet-Type Spaces D_α



- Smaller values of α correspond to larger spaces.
- Increasing α imposes stronger decay on the Taylor coefficients.

Cyclic Functions in D_α

$$D_\alpha = \left\{ f(z) = \sum_{n=0}^{\infty} a_n z^n : \sum_{n=0}^{\infty} (n+1)^\alpha |a_n|^2 < \infty \right\}$$

- $\alpha = 0$: f cyclic $\iff f$ outer.
- $\alpha = 1$: Brown–Shields conjecture (1984) :

$$f \text{ cyclic} \iff \begin{cases} f \text{ outer,} \\ c(\mathcal{Z}(f) \cap \mathbb{T}) = 0. \end{cases}$$

Here $c(\cdot)$ denote the logarithmic capacity.

- $\alpha > 1$: D_α is an algebra.

$$f \text{ cyclic} \iff \mathcal{Z}(f) \cap \overline{\mathbb{D}} = \emptyset.$$

Important Facts

- Since polynomials are dense in D_α , the constant function 1 is cyclic in D_α .
- If $g \in [f]$, then

$$[g] \subseteq [f].$$

Consequently, to prove that f is cyclic in D_α , it suffices to construct a sequence of polynomials

$$p_n \in \mathbb{C}[z]$$

such that

$$\|p_n f - 1\|_\alpha \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

- Suppose

$$f = gh,$$

where $g, h \in \mathbb{C}[z]$ are nonconstant polynomials.

Then

$$f \text{ is cyclic in } D_\alpha \iff g \text{ and } h \text{ are cyclic in } D_\alpha.$$

Some more results

A Necessary condition : Let $f \in D_\alpha(\mathbb{D})$ be cyclic. Then f has no zeros in $\mathbb{D}$.

A Sufficient condition : Let $f \in D_\alpha(\mathbb{D})$ be such that

$$\mathcal{Z}(f) \cap \overline{\mathbb{D}} = \emptyset.$$

Then f is cyclic in D_α .

Theorem (Brown & Shields)

Let $\zeta \in \mathbb{T}$. The polynomial $z - \zeta$ is cyclic in D_α if and only if $\alpha \leq 1$.

Cyclic Vector for a Commuting Pair of Operators

Let $T_1, T_2 \in \mathcal{B}(\mathcal{H})$ be commuting operators :

$$T_1 T_2 = T_2 T_1.$$

A vector $x \in \mathcal{H}$ is called **cyclic for** (T_1, T_2) if

$$\overline{\text{span}}\{T_1^{\alpha_1} T_2^{\alpha_2} x : (\alpha_1, \alpha_2) \in \mathbb{Z}_+^2\} = \mathcal{H}.$$

Equivalently,

$$\overline{\{p(T_1, T_2)x : p \in \mathbb{C}[z_1, z_2]\}} = \mathcal{H}.$$

That is, the smallest closed subspace invariant under both T_1 and T_2 containing x is $\mathcal{H}$.

Dirichlet-type spaces

For $\alpha \in \mathbb{R}$, consider

- the Dirichlet-type space denoted by $\mathfrak{D}_\alpha$ on $\mathbb{D}^2$, consisting of all holomorphic functions $f(z, w) = \sum_{k,l=0}^{\infty} a_{kl} z^k w^l$ such that

$$\|f\|_{\mathfrak{D}_\alpha}^2 := \sum_{k,l=0}^{\infty} (k+1)^\alpha (l+1)^\alpha |a_{kl}|^2 < \infty.$$

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$$\|f\|_\alpha^2 := \sum_{k,l=0}^{\infty} (k+l+1)^\alpha |a_{kl}|^2 < \infty.$$

Cyclic polynomials in $\mathfrak{D}_\alpha$

Theorem (Bénéteau–Khavinson–Kosiński–Liaw–Seco–Sola, *Trans. Amer. Math. Soc.*)

Let $p \in \mathbb{C}[z_1, z_2]$ be an irreducible polynomial with no zeros in the bidisk. We have the following :

- ① If $\alpha \leq \frac{1}{2}$, then p is cyclic in $\mathfrak{D}_\alpha$.
- ② If $\frac{1}{2} < \alpha \leq 1$, then p is cyclic in $\mathfrak{D}_\alpha$ if and only if
 $\mathcal{Z}(p) \cap \mathbb{T}^2$ is empty or finite,
or p is a constant multiple of $\zeta - z_1$ or of $\zeta - z_2$ for some $\zeta \in \mathbb{T}$.
- ③ If $\alpha > 1$, then p is cyclic in $\mathfrak{D}_\alpha$ if and only if $\mathcal{Z}(p) \cap \mathbb{T}^2$ is empty.

Question : Characterize cyclic polynomials in Dirichlet type space $\mathcal{D}_\alpha$?

Some partial answers are obtained by Torkinejad Ziarati and the following problem was asked in his recent paper :

Problem : Is the polynomial $2 - z_1 - z_2$ cyclic in $\mathcal{D}_\alpha$ for $\frac{3}{2} \leq \alpha \leq 2$?

Some comments

Recall that the boundary of $\mathbb{D}^2 = \mathbb{D} \times \mathbb{D}$ is

$$\partial(\mathbb{D}^2) = (\mathbb{T} \times \mathbb{D}) \cup (\mathbb{D} \times \mathbb{T}) \cup (\mathbb{T} \times \mathbb{T}).$$

Thus, zeros of a cyclic polynomial may occur on :

$$\mathbb{T}^2, \quad \mathbb{D} \times \mathbb{T}, \quad \mathbb{T} \times \mathbb{D}.$$

But if $p \in \mathbb{C}[z_1, z_2]$ is an irreducible polynomial with

- no zeros in $\mathbb{D}^2$,
- depends on both variables.

Then p cannot vanish on $\mathbb{D} \times \mathbb{T}$ and $\mathbb{T} \times \mathbb{D}$. This is an application of Hurwitz's theorem.

Cyclic polynomials in $\mathcal{D}_\alpha$

Theorem

Let $p \in \mathbb{C}[z_1, z_2]$ be an irreducible polynomial with no zeros in the bidisk. We have the following :

- ① *If $\alpha \leq 1$, then p is cyclic in $\mathcal{D}_\alpha$.*
- ② *If $1 < \alpha \leq 2$, then p is cyclic in $\mathcal{D}_\alpha$ if and only if $\mathcal{Z}(p) \cap \mathbb{T}^2$ is empty or finite.*
- ③ *If $\alpha > 2$, then p is cyclic in $\mathcal{D}_\alpha$ if and only if $\mathcal{Z}(p) \cap \mathbb{T}^2$ is empty.*

Model Polynomials

To understand cyclicity in $\mathcal{D}_\alpha$, it is natural to first investigate a few model polynomials.

The most important examples are :

$$1 - z_1, \quad 1 - z_1 z_2, \quad 2 - z_1 - z_2.$$

These polynomials represent the fundamental types of boundary zero sets on $\mathbb{T}^2$.

- $1 - z_1$:

$$\mathcal{Z}(p) \cap \partial(\mathbb{D}^2) = \{(1, z) : z \in \overline{\mathbb{D}}\}.$$

- $1 - z_1 z_2$:

$$\mathcal{Z}(p) \cap \mathbb{T}^2 = \{(e^{it}, e^{-it}) : t \in \mathbb{R}\},$$

a distinguished curve on $\mathbb{T}^2$.

- $2 - z_1 - z_2$:

$$\mathcal{Z}(p) \cap \mathbb{T}^2 = \{(1, 1)\},$$

an isolated boundary zero.

Thus these examples capture the main geometric possibilities for boundary zeros of polynomials on the bidisk.

A one variable polynomial

Let

$$p(z) = z - a, \quad |a| = 1,$$

and define

$$P(z_1, z_2) := p(z_1).$$

Then P is cyclic in $\mathcal{D}_\alpha$ if and only if

$$\alpha \leq 1.$$

Idea of the proof. For $f \in D_\alpha$, define

$$F(z_1, z_2) := f(z_1).$$

Then

$$\|f\|_{D_\alpha} = \|F\|_{\mathcal{D}_\alpha}.$$

Thus cyclicity of $P(z_1, z_2) = p(z_1)$ in $\mathcal{D}_\alpha$ reduces to cyclicity of p in the one-variable space D_α .

Using the Brown–Shields theorem :

$$p \text{ is cyclic in } D_\alpha \iff \alpha \leq 1.$$

Cyclicity of the polynomial $1 - z_1 z_2$

The polynomial

$$1 - z_1 z_2$$

is cyclic in $\mathcal{D}_\alpha$ if and only if

$$\alpha \leq 1.$$

Key observation.

If

$$F(z_1, z_2) = f(z_1 z_2), \quad \text{for some one variable function } f,$$

then for $\alpha > 0$,

$$\|f\|_{D_\alpha} \leq \|F\|_{\mathcal{D}_\alpha} \leq 2^\alpha \|f\|_{D_\alpha}.$$

Hence the map

$$f(z) \mapsto f(z_1 z_2)$$

preserves cyclicity up to norm equivalence.

Main Theorem

Theorem

Let $\alpha \leq 2$. Then

$$p(z_1, z_2) = 2 - z_1 - z_2$$

is cyclic in $\mathcal{D}_\alpha$.

Equivalently, if a sequence of complex numbers $\{b_{k,l}\}_{k,l \in \mathbb{Z}_+}$ satisfies

$$2b_{k,l} = b_{k+1,l} + b_{k,l+1}, \quad k, l \in \mathbb{Z}_+,$$

and

$$\sum_{k,l \in \mathbb{Z}_+} \frac{|b_{k,l}|^2}{(k+l+1)^2} < \infty,$$

then $b_{k,l} = 0$ for all $k, l \in \mathbb{Z}_+$.

Proof : Assume

$$f \in \mathcal{D}_2, \quad f \perp [p] \quad (\text{to prove : } f = 0).$$

Write

$$f(z_1, z_2) = \sum_{i,j=0}^{\infty} \frac{b_{ij}}{(i+j+1)^2} z_1^i z_2^j. \quad \left(\|f\|_{\mathcal{D}_2}^2 = \sum_{i,j=0}^{\infty} \frac{|b_{ij}|^2}{(i+j+1)^2} \right)$$

Since

$$f \perp z_1^k z_2^l (2 - z_1 - z_2),$$

we obtain

$$2b_{k,l} = b_{k+1,l} + b_{k,l+1}, \quad k, l \geq 0.$$

Define

$$g(z_1, z_2) := \sum_{i,j \geq 0} b_{ij} z_1^i z_2^j.$$

Since $f \in \mathcal{D}_2$, we have

$$g \in \mathcal{D}_{-2}.$$

Using the recurrence relation,

$$2b_{k,l} = b_{k+1,l} + b_{k,l+1},$$

we obtain

$$(z_1 + z_2 - 2z_1 z_2)g(z_1, z_2) = z_1 g(z_1, 0) + z_2 g(0, z_2).$$

We now introduce the substitutions

$$z_1 = \frac{\zeta}{\zeta - 1}, \quad z_2 = \frac{\zeta}{\zeta + 1}.$$

These satisfy :

$$z_1 \in \mathbb{D} \iff \Re \zeta < \frac{1}{2},$$

and

$$z_2 \in \mathbb{D} \iff \Re \zeta > -\frac{1}{2}.$$

Construction of an Entire Function

Substituting into the functional equation gives

$$\frac{\zeta}{\zeta-1}g\left(\frac{\zeta}{\zeta-1}, 0\right) + \frac{\zeta}{\zeta+1}g\left(0, \frac{\zeta}{\zeta+1}\right) = 0.$$

Define

$$h(\zeta) = \begin{cases} \frac{1}{\zeta-1}g\left(\frac{\zeta}{\zeta-1}, 0\right), & \Re\zeta < \frac{1}{2}, \\ -\frac{1}{\zeta+1}g\left(0, \frac{\zeta}{\zeta+1}\right), & \Re\zeta > -\frac{1}{2}. \end{cases}$$

Hence h is a well-defined entire function.

Using the fact that

$$g \in \mathcal{D}_{-2},$$

one obtains

$$\int_{|\zeta|>1} \frac{|h(\zeta)|^2}{|\zeta|^4} dA(\zeta) < \infty.$$

This implies that h is a polynomial of degree at most 1 :

$$h(\zeta) = a(\zeta - 1) + b.$$

A further integrability argument shows that

$$a = 0.$$

Hence

$$h(\zeta) = b.$$

Using the definition of h ,

$$g(z_1, 0) = \frac{b}{z_1 - 1}, \quad g(0, z_2) = \frac{b}{z_2 - 1}.$$

Substituting into the functional equation yields

$$(z_1 + z_2 - 2z_1z_2)g(z_1, z_2) = \frac{bz_1}{z_1 - 1} + \frac{bz_2}{z_2 - 1}.$$

Solving,

$$g(z_1, z_2) = -\frac{b}{(1 - z_1)(1 - z_2)}.$$

Using the geometric series expansion,

$$g(z_1, z_2) = -b \sum_{k, l \geq 0} z_1^k z_2^l.$$

Hence

$$a_{kl} = -b, \quad k, l \geq 0.$$

Conclusion

Now compute

$$\|g\|_{-2}^2 = |b|^2 \sum_{k,l \geq 0} \frac{1}{(k+l+1)^2}.$$

Grouping terms according to $n = k + l$,

$$\sum_{k,l \geq 0} \frac{1}{(k+l+1)^2} = \sum_{n=0}^{\infty} \frac{n+1}{(n+1)^2} = \sum_{n=0}^{\infty} \frac{1}{n+1}.$$

This series diverges.

Therefore,

$$g \notin \mathcal{D}_{-2},$$

unless

$$b = 0.$$

Hence

$$g = 0, \quad f = 0.$$

Thus

$$[p]^\perp = \{0\},$$

and therefore

$$p(z_1, z_2) = 2 - z_1 - z_2$$

is cyclic in $\mathcal{D}_\alpha$ for all $\alpha \leq 2$.

Lemma

Let $f \in \mathbb{C}[z_1, z_2]$. Assume that

- f has no zeros in $\mathbb{D}^2$
- f has finitely many zeros on $\mathbb{T}^2$, i.e.,

$$\mathcal{Z}(f) \cap \mathbb{T}^2 = \{(\zeta_j, \eta_j) \in \mathbb{T}^2, j = 1, \dots, k\}.$$

Then for any integer m there exists sufficiently large N such that the function

$$Q(z_1, z_2) = \frac{\prod_{i=1}^k (2 - \zeta_i^{-1} z_1 - \eta_i^{-1} z_2)^N}{f(z_1, z_2)}$$

is m -times differentiable on $\mathbb{T}^2$.

Polynomials with Finitely Many Boundary Zeros

Theorem

Let $p \in \mathbb{C}[z_1, z_2]$ be a polynomial having

- no zeros in $\mathbb{D}^2$
- only finitely many zeros on $\mathbb{T}^2$.

Then p is cyclic in $\mathcal{D}_\alpha$ for every

$$\alpha \leq 2.$$

Proof.

There exists a function

$$Q(z_1, z_2) = \frac{\left(\prod_{i=1}^k (2 - \zeta_i^{-1} z_1 - \eta_i^{-1} z_2)\right)^N}{p(z_1, z_2)} = \frac{g(z_1, z_2)}{p(z_1, z_2)},$$

where $(\zeta_i, \eta_i) \in \mathbb{T}^2$, such that Q is twice continuously differentiable on $\mathbb{T}^2$.

Hence its Fourier coefficients satisfy

$$\sum_{k,l} |\hat{Q}(k, l)|^2 (k+1)^2 (l+1)^2 < \infty.$$

Since

$$(k + l + 1)^2 \leq (k + 1)^2(l + 1)^2,$$

we obtain

$$\sum_{k,l} |\hat{Q}(k, l)|^2 (k + l + 1)^2 < \infty.$$

Therefore,

$$Q \in \mathcal{D}_\alpha, \quad \alpha \leq 2.$$

Hence

$$g(z_1, z_2) \in p\mathcal{D}_\alpha.$$

On the other hand,

$$g(z_1, z_2) = \left(\prod_{i=1}^k (2 - \zeta_i^{-1} z_1 - \eta_i^{-1} z_2) \right)^N$$

is cyclic in $\mathcal{D}_2$.

Since p is a multiplier of $\mathcal{D}_2$, it follows that p is cyclic in $\mathcal{D}_2$, and consequently in $\mathcal{D}_\alpha$ for all $\alpha \leq 2$.

Question

For $\vec{\alpha} = (\alpha_1, \alpha_2, \alpha_3) \in \mathbb{R}^3$, consider the Dirichlet-type space denoted by $\mathcal{D}_{\vec{\alpha}}$ on $\mathbb{D}^2$, consisting of all holomorphic functions

$$f(z, w) = \sum_{k,l=0}^{\infty} a_{kl} z^k w^l$$

such that

$$\|f\|_{\mathcal{D}_{\vec{\alpha}}}^2 := \sum_{k,l=0}^{\infty} (k+1)^{\alpha_1} (l+1)^{\alpha_2} (k+l+1)^{\alpha_3} |a_{kl}|^2 < \infty.$$

Question : Characterize cyclic polynomial in $\mathcal{D}_{\vec{\alpha}}$.

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Thank You For Your Attention !